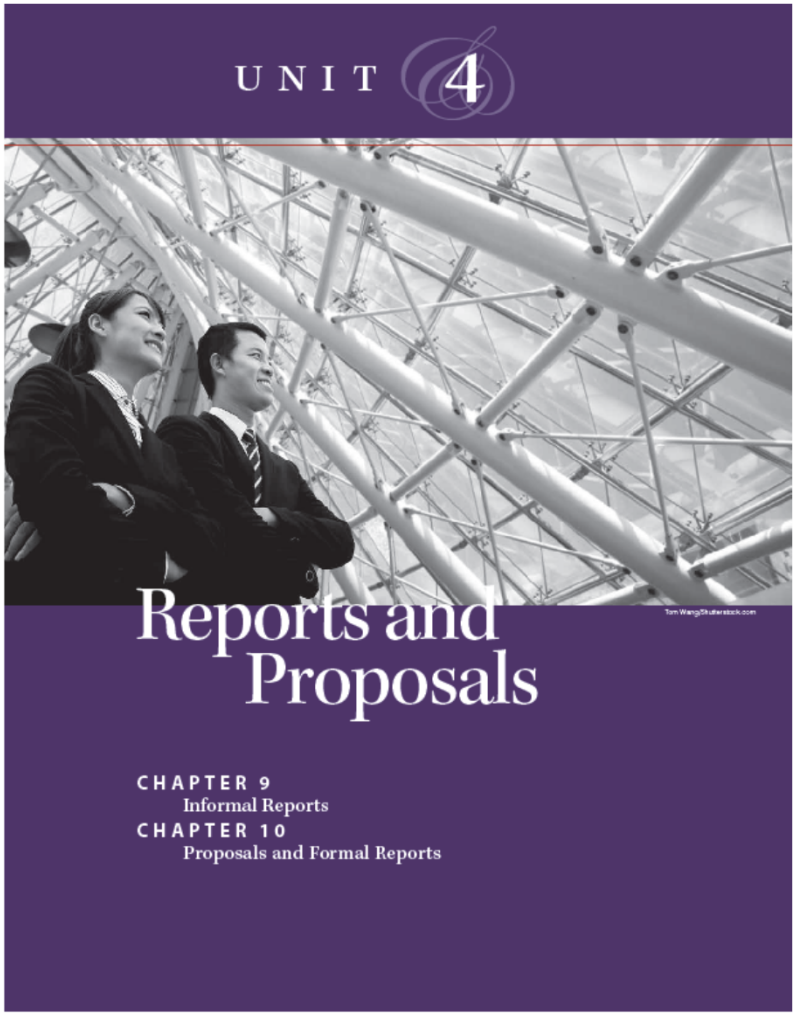
# Fall 2019 VE496
# Lecture notes
# Week 3

# Creating a Professional Portfolio

A portfolio is a collection of materials that you can use to demonstrate your qualifications and abilities.

- CV/Resume
- Samples of written work (Statement of Purpose)
- Examples of presentations (remote interview)
- Descriptions and evidence of projects
- Diplomas and certificates & awards
- Letters of reference

# Reading for Week 3, 5, 6



# Letters, Memos, and E-Mails

**Letters** are written to people ***outside*** the company or organization.

Primarily used in formal situations where:

- an employee is acting as a representative of the company to make requests or inquiries, communicate important information, record agreements
- a person is applying for jobs

**Memos** are written to people ***inside*** the company or organization.

- Usually contain meeting agendas, policies, internal reports, and short proposals.
- Memos more reliable than e-mails for info that should not be broadly released.

**E-mails** can be written to people ***inside* or *outside*** the company or organization.

-- E-mail is used in situations that once called for less formal memos, letters, or phone calls. Increasingly, e-mail is being used for formal communication, too.

## Basic Features of Letters, Memos, and E-mails

A letter, memo, or e-mail will generally have the following features:

- Salutation (*for letters and some e-mails but not memos*)
- Introduction that states a clear main point
- Body paragraphs with need-to-know information
- Conclusion that restates the main point
- Signature (*for letters and e-mails but not memos*)

# Proposals & Reports

## Proposals

Offer structured persuasion for internal or external audience

## Informational Reports

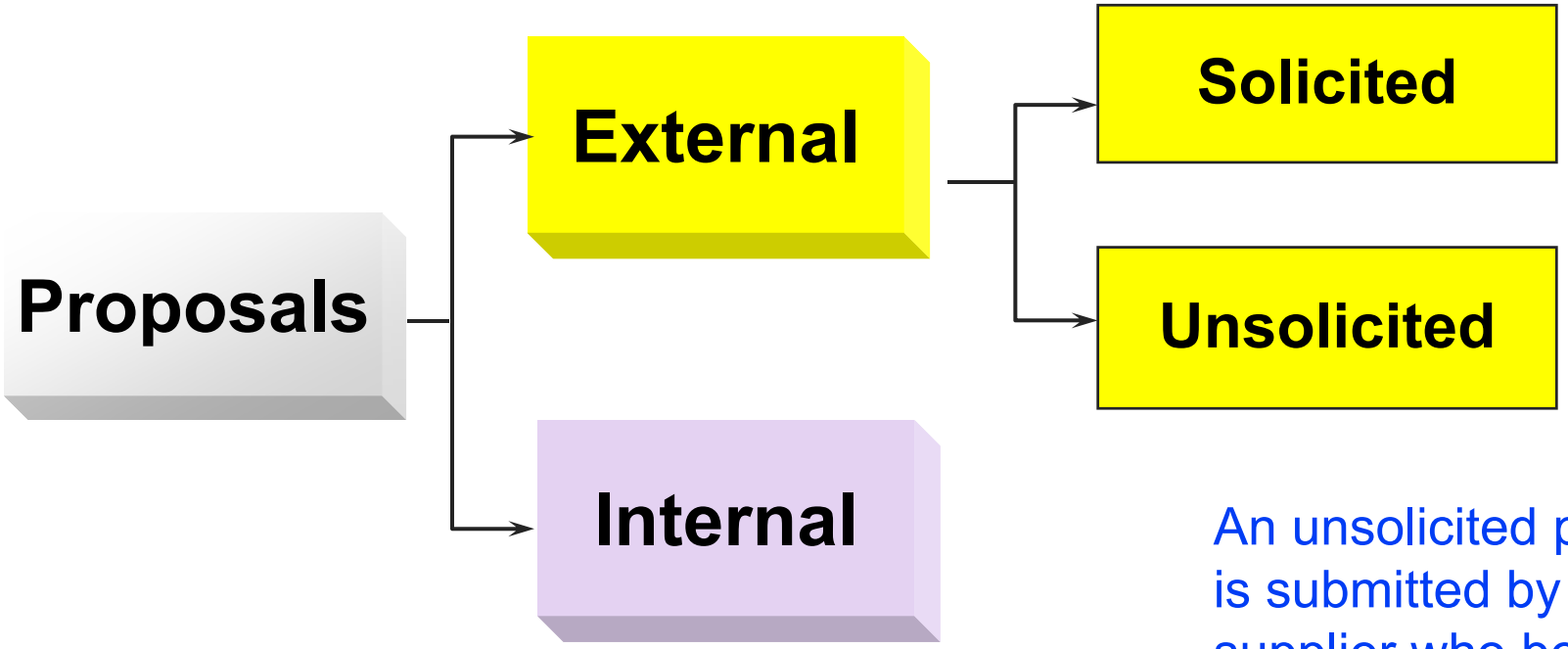
Offer available data, facts, without analysis or recommendation

## Analytical reports

Offer both info & analysis, and they can also include recommendation

External proposals are submitted to another organization.

A solicited proposal is submitted to a Request for Proposal (RFP) from a prospective customer (company or government)




An unsolicited proposal is submitted by a supplier who believes the prospective customer has a need for goods or services.

Internal proposals are submitted to the writer's own organization

# Evaluation criteria for competitive proposals

- **Relevance:** What problem are you going to solve?
- **Innovation:** How are you going to solve it?
- **Deliverable:** What exactly will you provide for us?
- **Feasibility:** Can you deliver what you promise?
- **User benefits:** What benefits can you offer?
- **Timeline:** When will you complete the work?
- **Cost:** How much will you charge?

**A proposal is an act of persuasion.**

**It is a sales document.**

# Cost-benefit analysis (CBA)

CBA has two purposes:

1. To determine if an investment/decision is sound (justification/feasibility) – verifying whether its benefits outweigh the costs, and by how much;

2. To provide a basis for comparing projects – which involves comparing the total expected cost of each option against its total expected benefits.

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## Proposals have 2 goals

1. To get the project accepted
2. To get you accepted to do the project

   1 + 2 = the winning bid

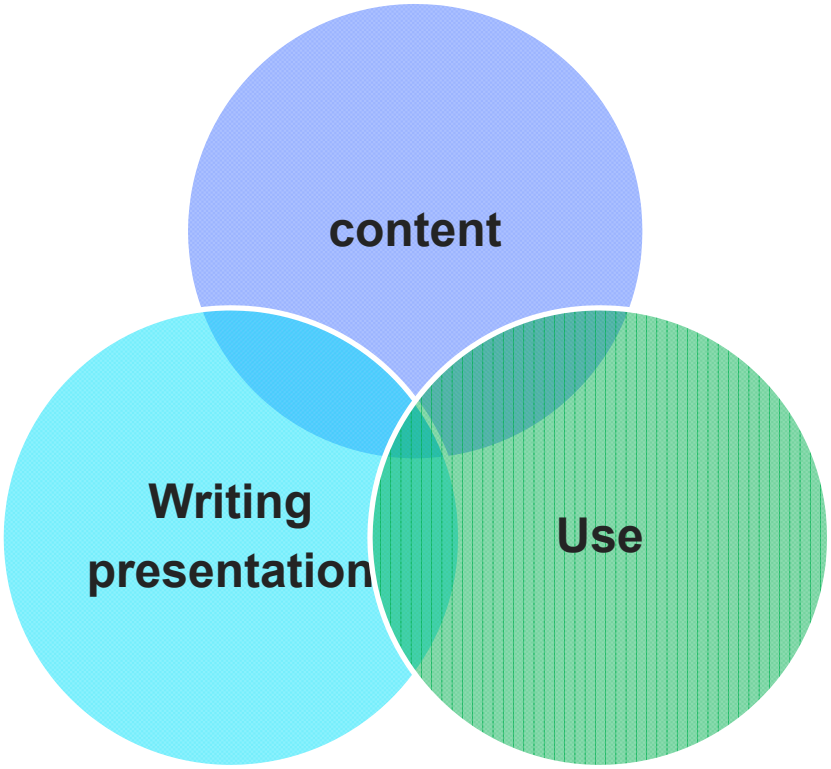
## Must stress:

- Reader benefits
- Specific supporting evidence

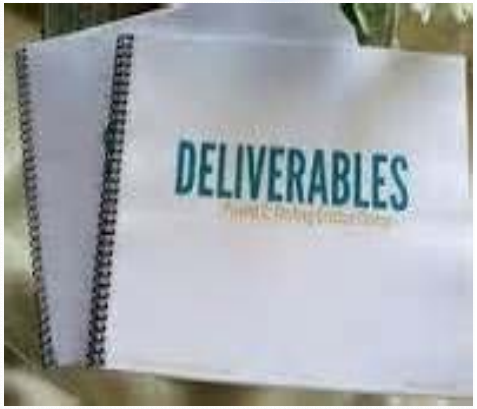
## Must present professional image:

- Quality of content
- Quality of presentation

# Content and form CANNOT be separated

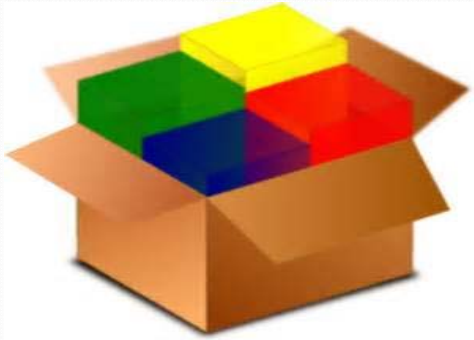


# Deliverables




**Two major categories:**

1. **Research**
2. **Goods (**products**) and/or services (**training or counsel**)**

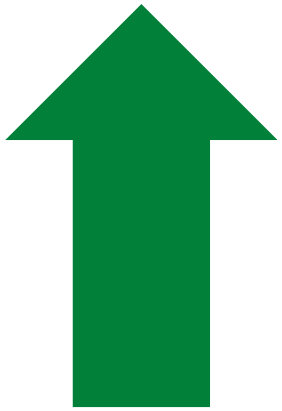


# What is a report?

**A variety of reports**:

- A document that contains researched information and numerical data
- Presentation slides or Excel sheets printed out

- **Formal** reports contain a title page, a table of contents, a list of illustrations, appendix, etc.
- **Informal** reports may be letters or memos distributed by email



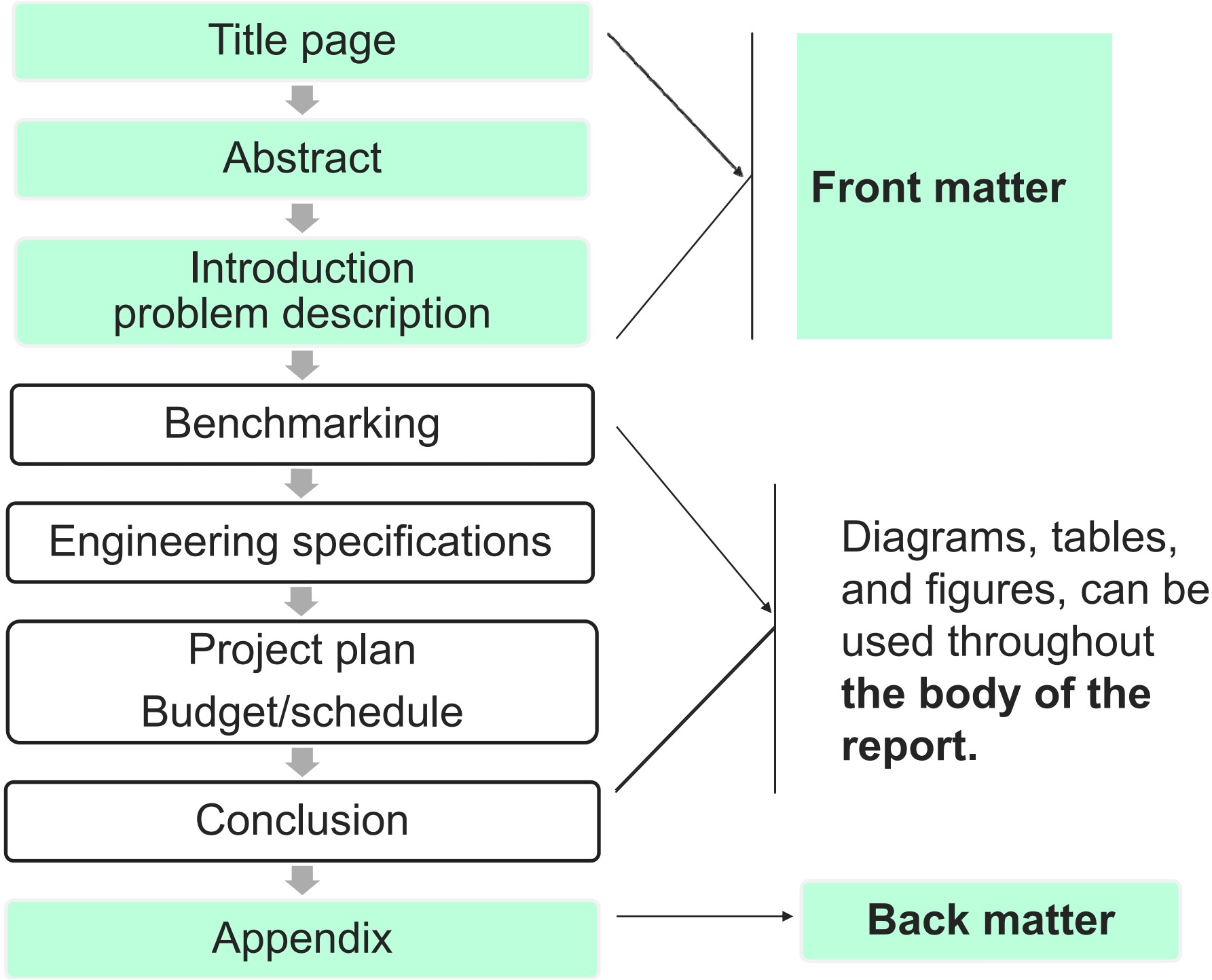
Degree pf formality

“Front matters”

“Back matters”



Informality


Title page
Abstract
Introduction
problem description
Benchmarking
Engineering specifications
Project plan
Budget/schedule
Conclusion
Appendix
Front matter
Diagrams, tables, and figures, can be used throughout **the body of the report.**
Back matter

# 3 levels of reports

## Information report: (orderly info)

A report listing all the competition within a 5-mile radius, grouped according to location or type of company

## Analytical report ( info+)

A report listing all the competition within a 5-mile radius **AND** draws conclusions about the likely demand for a new hardware store in the area.

## Recommendation report (info++)

A report listing all the competition within a 5-mile radius **AND** draws conclusion about the likely demand for a new hardware store in the area **AND** recommends a decision or further action.

## Organizing Informational Reports:

- **Comparison.** Show similarities and differences (or advantages and disadvantages) between two or more entities
- **Importance.** Build up from the least important item to the most important (or from most important to the least, if you don't think your audience will read the entire report)
- **Sequence.** Organize the steps or stages in a process or procedure
- **Chronology.** Organize a chain of events in order from oldest to newest or vice versa
- **Geography.** Organize by region, city, state, country, or other geographic unit
- **Category.** Group by topical category, such as sales, profit, cost, or investment

# Analytical reports:

- **Reports to assess opportunities.**

Every business opportunity carries some degree of risk and also requires a variety of decisions and actions

You can use analytical reports to **assess the risks involved.**

For instance,
- ***Market analysis reports*** are used to judge the likelihood of success for new products or sales.

***- Due diligence reports*** examine the financial aspects of a proposed decision, such as acquiring another company.

# Analytical reports:

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- **Reports to solve problems.**

Managers often ask for ***troubleshooting reports*** when they need to understand why something isn't working properly and what needs to be done to fix it.

***Failure analysis report*****,** studies events that happened in the past, with the hope of learning how to avoid similar failures in the future (also known as the ***post mortem report***).

# Analytical reports:

- **Reports to support decisions.**

  ***Feasibility reports*** are called for when managers need to explore the ramifications of a decision they're about to make, such as switching materials used in a manufacturing process.

  ***Justification reports*** explain a decision that has already been made.

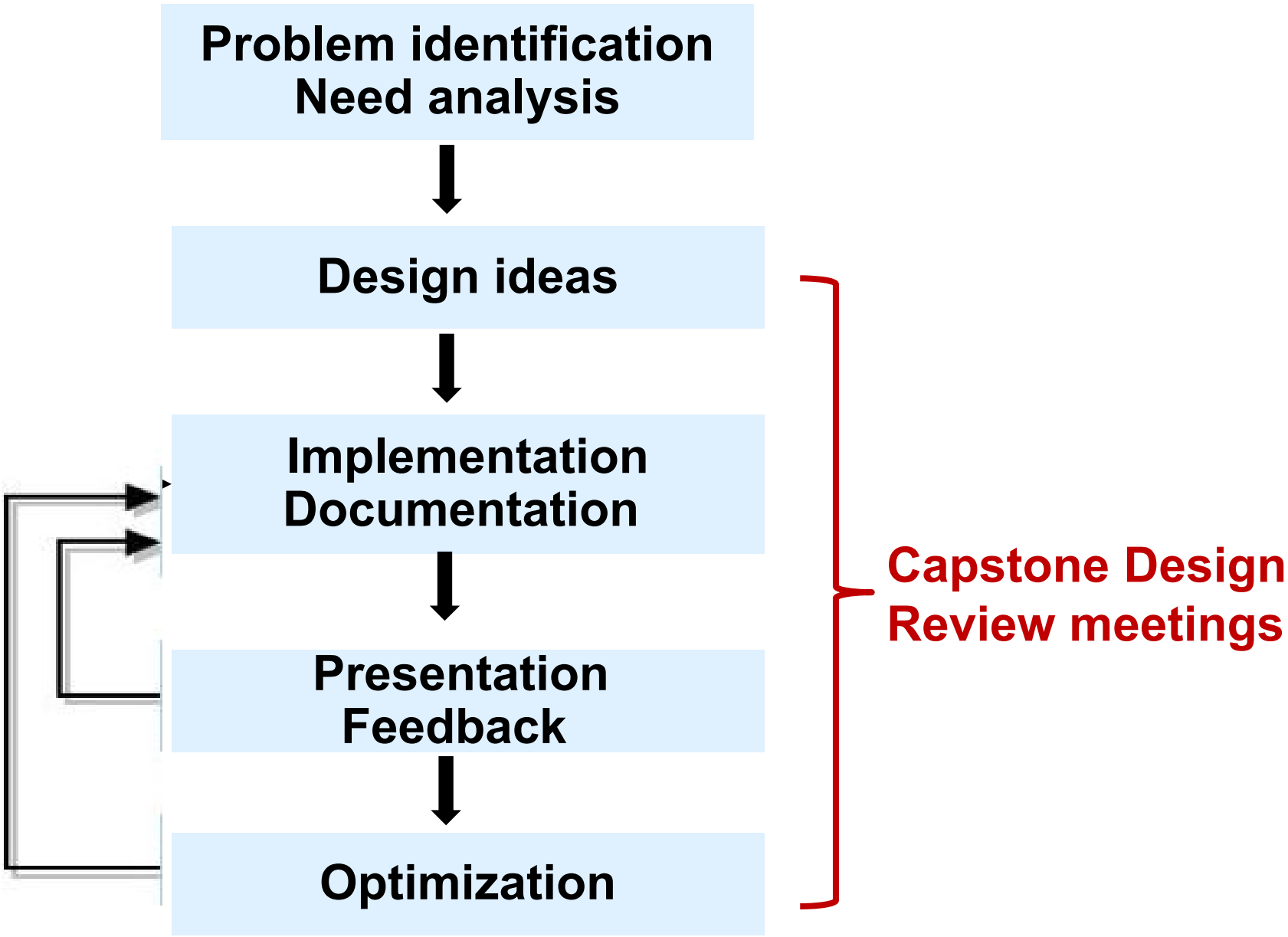
# Recommendation reports:

- **Reports to provide recommendations.**
  - Workplace research papers carried out to clarify short-, mid-, and long- term business development interests.
  - Often prepared by *internal employees* and *external consultants.*

## Need/problem vs. purpose

Someone (usually your supervisor) needs information for a research purpose and has authorized you to do the work.

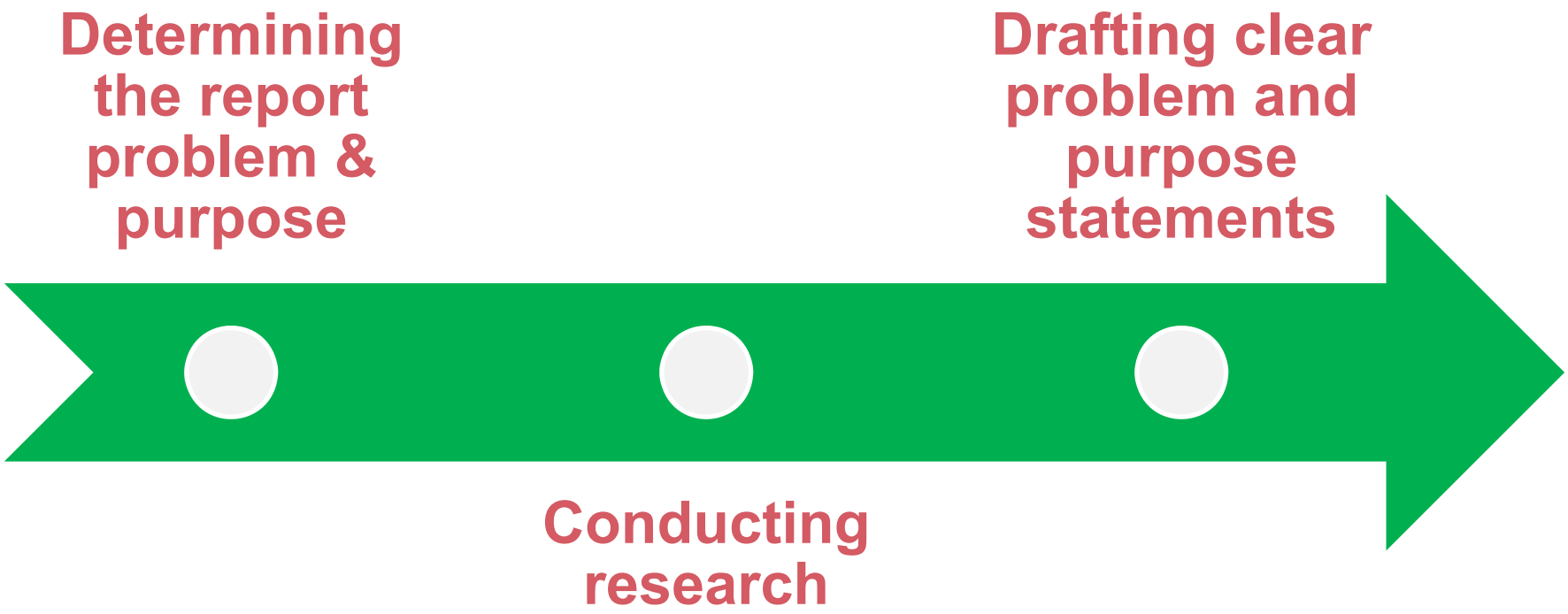
How you define this need (problem) will determine your report's purpose.


Problem identification
Need analysis
Design ideas
Implementation
Documentation
Presentation
Feedback
Optimization
Capstone Design
Review meetings

## Ask yourself:

- ***What has my reader asked for?***
- ***What will my data support*?**

**Then deliver *exactly* that, no more, no less.**


Determining the report problem & purpose
Conducting research
Drafting clear problem and purpose statements

## Primary research

- Uncovers information firsthand
- Produces new information through the use of experiments, surveys, interviews, & other methods of direct observation.

## Secondary research

- Uses material that others have published in periodicals, brochures, books, digital publications, and Web sites.
- Typically conducted **before** you engage in primary research.

# Writing literature review

- **surveys** the existing scholarly research on a particular topic.
- **gives an overview** of key findings, concepts and developments in relation to a research problem or question
- **may be presented** as a standalone paper or as a portion of a larger research project.

**A good literature review aims to:**

- Analyze and critically evaluate the literature
- Synthesize sources to highlight:
  - patterns
  - trends
  - points of consensus
  - points disagreement
  - research gaps
  - open questions that remain

## Why Write a Literature Review (IR)?

1. **Justifying your own research**. IR allows you to demonstrate what makes your own research valuable. It should serves as a jumping-off point for the rest of your own research.

2. **Demonstrating your expertise.** A well done IR shows that you are able to synthesize and logically present the insights gained. The final product establishes you as a trustworthy authority on your topic.

3. **Joining the conversation**. By producing a literature review, you're engaging with all of the prior scholars who examined your topic and continuing a cycle that moves the field forward.

## 3 Strategies for Writing a Literature Review:

1. **Organizing:**
   - identifying the key articles or the key areas of knowledge
   - associating papers with specific areas of knowledge.

2. **Summarizing:**
   - identifying each article's contribution to the area of knowledge by asking the following questions:

   a) What is the author's purpose?
   b) What are the writer's assumptions?
   c) What are the author's main claims (conclusions)? How are they supported; how have they been qualified?

3. **Evaluating:** In the final step, you need to assess the work done in the key area of knowledge in order to establish:

- **How** previous work has left a gap, because of either inadequate assumptions or inconclusive findings
- **How** previous research will be applied in a new context
- **How** general disagreement or different views on the subject create a need for a solution

In evaluating several papers on a specific topic, ask these questions:

- **What** are the significant points of agreement between articles?
- **Where** the research disagrees, is one researcher more conclusive than another?
- **How** can you fit the articles together to build a logical argument that furthers your purpose?

## Writing a Problem Statement:

- Provides a clear description of the situation that created the need for your report.

- Generally written as a declarative statement.

Example:

"*Sales are declining at Company X.*"

Writing a **Purpose Statement:**

(also called the report's **objective**, **aim**, or **goal**).

It is often written in the form of:

1. **a research question**
2. **infinitive phrase** ("**to…..**")

**Example:**

If your problem is that Company X wants to know why sales are declining, your purpose statement may be:

- "What are the causes of decreasing sales at X?"
- "to determine the causes of decreasing sales at X"

# Thank you!

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